MERMAID: A multi-armed bandit approach for optimal carrier selection in OFDM-based underwater acoustic networks

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***Abstract*—Reliable communication in UnderWater (UW) environments is hindered by limited bandwidth, high temporal variability, and significant propagation delays. Orthogonal Frequency Division Multiplexing (OFDM) has emerged as a promising physical-layer technique for UW communication; however, its performance depends heavily on the dynamic selection of active subcarriers to track channel fluctuations. This paper presents MERMAID, the first framework that leverages a Combinatorial Multi-Armed Bandit (CMAB) approach for adaptive subcarrier selection in Underwater Acoustic Networks (UANs). MERMAID dynamically adjusts the number of active subcarriers based on real-time Packet Delivery Ratio (PDR) feedback, aiming to maximize PDR while minimizing energy consumption. Unlike Deep Reinforcement Learning (DRL) methods—which require extensive training and incur high computational overhead—MERMAID is designed for resource-constrained UW nodes, enabling fast adaptation and lightweight operation. We implement MERMAID in the DESERT simulator and benchmark its performance against state-of-the-art baselines. Simulation results show that MERMAID improves PDR by up to 21.15% and reduces energy consumption by up to 45.40%. These results confirm MERMAID as a lightweight, scalable, and high-performance solution for real-world Internet of Underwater Things (IoUT) deployments.**

***Index Terms*—Underwater acoustic networks, OFDM, subcarrier selection, multi-armed bandits, energy efficiency**

I. Introduction

Acoustic communication is the most widely adopted technology for UnderWater (UW) networks, offering superior range and reliability [1], [2]. However, UW systems face critical limitations, including high temporal variability, limited bandwidth, low data rates, and significant propagation delays [3]. These constraints underscore the need for dynamic optimization of transmission parameters to ensure communication efficiency in such harsh and variable environments.

Orthogonal Frequency Division Multiplexing (OFDM) is one of the most promising technologies for achieving high data rates in UW communications. By enabling parallel transmission over multiple subcarriers, OFDM effectively mitigates Inter-Symbol Interference (ISI) and improves spectral efficiency. However, UW Acoustic (UWA) channels are inherently dynamic due to frequent fluctuations in channel quality, severe multipath propagation, and Doppler-induced distortions. Under such conditions, a fixed subcarrier allocation becomes suboptimal, resulting in inefficient bandwidth utilization. To overcome this limitation, adaptive subcarrier selection—i.e., dynamically selecting *which* subcarriers to use based on real-time channel feedback—emerges as a key strategy to enhance overall system performance.

Recent studies have explored various techniques to enhance OFDM-based UW communication. For instance, [4] introduces Orthogonal Chirp-based Frequency Division Multiplexing (OCFDM) to reduce transmission time and improve robustness against channel impairments. In [5], the authors investigate multifrequency OFDM schemes tailored for hydroacoustic environments, showing that constant-envelope signals offer greater reliability in practice. Other works, such as [6], propose designs based on the Fractional Fourier Transform (FrFT) to reduce equalization complexity and enhance feasibility in constrained UW wireless systems. Collectively, these contributions highlight the potential of adaptive OFDM strategies to address the challenges of dynamic UW channels. Nonetheless, the problem of real-time subcarrier selection under stringent computational and energy constraints remains insufficiently investigated.

While model-based methods can determine the optimal number of subcarriers, they often fail to generalize across the diverse and unpredictable conditions of UW environments. Variations in depth, salinity, temperature gradients, and seabed morphology significantly affect acoustic channel characteristics [7], [8], rendering static or analytically derived models inadequate. As an alternative, Machine Learning (ML) techniques have shown promise in dynamically optimizing communication parameters based on real-time observations.

However, traditional Reinforcement Learning (RL) and Deep RL (DRL) techniques are often ill-suited for UW systems for two main reasons. First, UW nodes are highly resource-constrained, with limited processing power and memory, making complex learning algorithms impractical. Second, these methods typically require long convergence times, which are incompatible with rapidly varying UW channel conditions

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where swift adaptation is essential. These limitations call for lightweight, fast-converging learning frameworks that can adapt to dynamic environments while remaining computationally feasible for deployment on UW nodes [9]–[11].

Given these limitations, Multi-Armed Bandits (MABs) emerge as a compelling alternative to conventional RL techniques for UW communications [12]–[14]. MABs operate without explicit state modeling or lengthy training, making them inherently lightweight, adaptive, and well-suited to resource-constrained scenarios. Motivated by these advantages, we propose MERMAID (Multi-armEd bandit approach foR optiMal cArrier selection In OFDM-based unDerwater acoustic networks), a novel framework that employs Combinatorial MAB (CMAB) learning to optimally select the best OFDM subcarriers based on real-time feedback on Packet Delivery Ratio (PDR).

The main contributions of this work are as follows:

- We introduce MERMAID, the first CMAB-based framework for adaptive subcarrier selection in UWA networks. Designed for resource-constrained Internet of Underwater Things (IoUT) nodes, MERMAID provides a lightweight, fast-converging solution to maximize PDR and energy efficiency in variable UW environments.
- We implement MERMAID in the DESERT simulator [15], [16], a widely adopted platform for UWA network evaluation, and benchmark its performance against state-of-the-art DRL baselines under realistic UW channel conditions.
- We show that, despite its low computational complexity, MERMAID converges faster and delivers more stable performance than DRL-based approaches. It achieves **up to 21.15% higher PDR and 45.40% energy savings**, enabling real-time adaptation under the tight resource constraints of UWA nodes. These gains stem from MERMAID's principled exploration and rapid exploitation of subcarrier configurations aligned with current channel conditions, avoiding the overhead and instability typical of neural policy learning.

The remainder of this paper is organized as follows. Section II outlines the system model and network architecture. Section III presents the fundamentals of the CMAB algorithm. Section IV details the performance evaluation and comparison with state-of-the-art baselines. Finally, Section V draws some conclusions on this work.

II. System Model

The proposed network architecture, illustrated in Fig. 1, features a single IoUT node which collects sensor data of various nature and forwards them to a central receiver node (e.g., a Sink node) via an acoustic OFDM link. The latter aggregates the received data and forwards it through a wired connection to a *Supporting Vessel*, which provides continuous power and external connectivity to the receiver.

In particular, the available bandwidth is divided into N uniformly spaced OFDM subcarriers. The transmitter node, in turn, can select any subset of $K < N$ subcarriers to send the data.

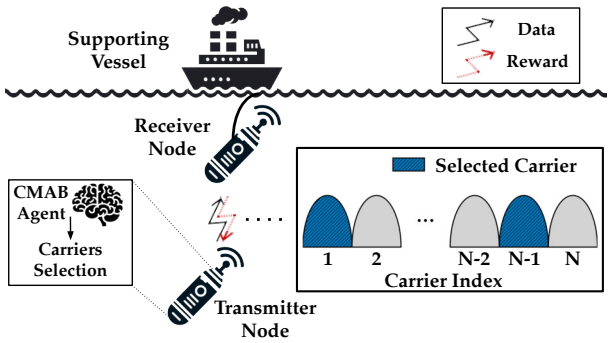


Fig. 1: System architecture.

To this end, the transmitter node hosts a local CMAB agent, tasked with selecting the K OFDM subcarriers to maximize PDR and energy efficiency of the UW OFDM communication.

In this view, MERMAID follows a structured decision-making process, organized into discrete *episodes*. Each episode spans a fixed time interval Δ, during which the agent selects a configuration of active OFDM subcarriers. The transmitter then applies the selected configuration, updating its parameters accordingly. Over successive episodes, the agent refines its subcarrier selection strategy based on accumulated feedback, progressively improving communication performance.

At each episode n, the agent selects the K OFDM subcarriers to employ for transmission. Afterwards, it receives a *reward* reflecting the achieved PDR[1]. This feedback guides future decisions, allowing the agent to iteratively refine its subcarrier selection policy.

Let P_n^r denote the number of packets successfully received at the receiver node during the interval between episodes $n-1$ and n, computed as:

$$P_n^r = \sum_{t \in \mathcal{T}_n} p_t^r, \tag{1}$$

where $\mathcal{T}_n$ is the set of transmission attempts in $[(n-1)\Delta,\, n\Delta)$, and p_t^r is defined as:

$$p_t^r = \begin{cases} 1 & \text{if a packet is successfully received at time } t, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Since the goal is to maximize reliability, the *reward function* at episode n is equal to the PDR experienced within $[(n-1)\Delta,\, n\Delta)$:

$$r_n = \frac{P_n^r}{|\mathcal{T}_n|}, \tag{3}$$

where $\frac{P_n^r}{|\mathcal{T}_n|}$ is the ratio of successfully received packets to transmitted packets in episode n.

[1]Since the receiving node is continuously powered and capable of high-power transmission, we assume that receiver feedback is reliably received regardless of the selected subcarriers.

III. Combinatorial Multi-Armed Bandit Algorithm

In this section, we illustrate how the MERMAID agent dynamically selects the optimal set of active OFDM subcarriers using a Combinatorial Multi-Armed Bandit (CMAB) framework [17], [18].

MAB algorithms are widely used in online learning, particularly for resource allocation in wireless networks. In the classical setting, the agent selects a single action (or *arm*) at each episode. The CMAB framework generalizes this idea by allowing the selection of a subset of actions—called a *super arm*—from a finite ground set. This extension captures interdependencies among arms and expands the decision space, enabling more effective learning in complex, dynamic scenarios such as UWA systems.

In our setting, each arm corresponds to a single OFDM subcarrier, while a super arm is a subset of subcarriers activated during an episode. The agent aims to select, at each episode, the subset that maximizes the communication PDR. The key challenge lies in the combinatorial nature of the action space: the number of possible super arms grows exponentially with the size of the subcarrier set.

The expected reward for a given subcarrier subset $\mathcal{A}_n$ selected at episode n is defined as:

$$Q(\mathcal{A}_n) = \mathbb{E}[r_n \mid \mathcal{A}_n], \tag{4}$$

where $\mathbb{E}[\cdot]$ denotes the expectation over the channel dynamics, and r_n is the reward observed at episode n after selecting $\mathcal{A}_n$.

To make effective decisions, the agent must balance *exploration*—testing previously untried subcarrier subsets—and *exploitation*—choosing the best-known configuration to maximize long-term performance.

A. Upper Confidence Bound for CMAB

To efficiently explore the combinatorial action space, we adopt a Combinatorial Upper Confidence Bound (UCB)-based strategy, extending the classical UCB algorithm [19] to the CMAB setting. UCB prioritizes actions with high estimation uncertainty, allowing the agent to refine reward predictions over time. In our context, it balances empirical performance and exploration by favoring subcarrier subsets with greater uncertainty, ensuring systematic and efficient learning.

The optimal subset of active subcarriers at episode n is selected as:

$$\mathcal{A}_n = \arg\max \mathcal{A}_i \subseteq \mathcal{S}\left(Q_n(\mathcal{A}_i) + \sum_{s \in \mathcal{A}_i} \sqrt{\frac{c \cdot \log n}{N_n(s)}} \right), \tag{5}$$

where $Q_n(\mathcal{A}_i)$ is the empirical mean reward of subset $\mathcal{A}_i$ up to episode n, and $N_n(s)$ is the number of times subcarrier s was selected. The parameter $c > 0$ tunes the exploration–exploitation balance. Unlike standard MABs, the total uncertainty is computed by summing confidence bounds across all subcarriers in the subset, promoting exploration of underused combinations.

The mean reward estimate $Q_n(\mathcal{A}_i)$ is computed as:

$$Q_n(\mathcal{A}_i) = \frac{\sum_{k=1}^{n} r_k \cdot \mathbb{1}\{\mathcal{A}_k = \mathcal{A}_i\}}{\sum_{k=1}^{n} \mathbb{1}\{\mathcal{A}_k = \mathcal{A}_i\}}, \tag{6}$$

where r_k is the reward observed at episode k, and $\mathbb{1}\{\mathcal{A}_k = \mathcal{A}_i\}$ is an indicator function equal to 1 if subset $\mathcal{A}_i$ was selected at episode k, and 0 otherwise.

By capturing both individual subcarrier performance and their interactions, the CMAB framework allows the agent to refine its strategy over time. Initially, the algorithm favors subsets with underexplored subcarriers to ensure broad exploration. As reward estimates become more reliable, the policy shifts toward exploitation, consistently selecting high-performing combinations. This adaptive behavior enables robust performance even under dynamic and uncertain UW channel conditions.

IV. Performance Evaluation

We evaluate MERMAID's performance through extensive simulations using the DESERT simulator. The environment reproduces realistic UWA conditions, including frequency-dependent attenuation, ambient noise from turbulence and vessels, and Doppler-induced variability, providing a faithful model of practical UW impairments.

To assess MERMAID's effectiveness, we benchmark it against two widely adopted DRL algorithms: Double Deep Q-Network (DDQN) [20] and Proximal Policy Optimization (PPO) [21]. Both baselines are adapted to address the same subcarrier selection task and operate under identical environmental and feedback conditions, ensuring a fair comparison. Specifically:

- DDQN is a value-based DRL algorithm that addresses the overestimation bias of standard Q-learning by decoupling action selection from evaluation via a target network. In our setup, it maps link statistics to the optimal number of active OFDM subcarriers, updating Q-values through experience replay.
- PPO is a policy-gradient algorithm that enhances training stability using clipped surrogate objectives and adaptive trust region constraints. It learns a stochastic policy for subcarrier selection, optimized via gradient ascent to maximize long-term expected reward.

Both agents rely on ground-truth feedback from the simulator and adopt stateful learning paradigms based on full observability and prolonged training. While effective in controlled settings, they demand significant computational resources and long convergence times, limiting their suitability for real-time use on resource-constrained IoUT devices.

In contrast, MERMAID employs a stateless CMAB-based approach that ensures rapid convergence, low memory usage, and real-time adaptability, without requiring prior training.

A. Simulation Setup

We evaluated MERMAID's performance in the point-to-point UWA link described in Section II, where, as already mentioned, a single transmitting node communicates with a

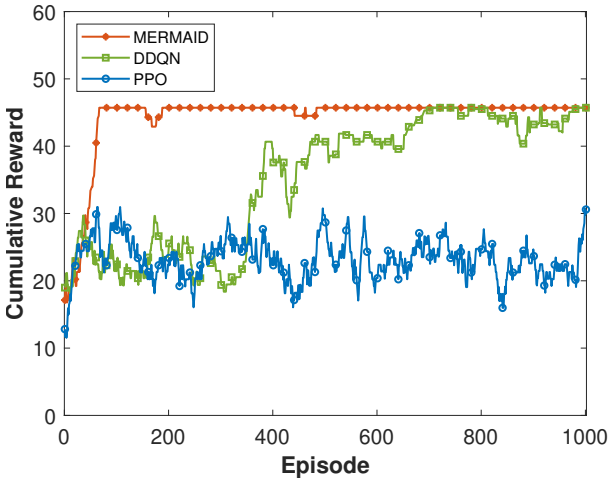


Fig. 2: Cumulative reward comparison between MERMAID, DDQN and PPO.

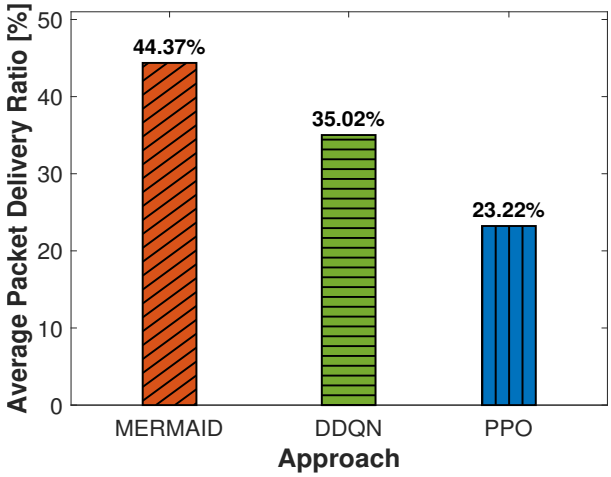


Fig. 3: Average PDR comparison between MERMAID, DDQN and PPO.

central receiver over a direct acoustic channel. This setup isolates subcarrier adaptation from multi-hop interference and routing dynamics.

The simulated topology consists of a single transmitter–receiver pair, with the receiver node fixed and the transmitter located 360 meters away. This setup reflects typical IoUT scenarios characterized by sparse deployments and long communication ranges. The chosen distance emulates realistic acoustic propagation, accounting for attenuation and time-varying multipath effects [22].

The physical transmission parameters are based on real-world UWA modems, such as EvoLogics devices [23]. The system operates with a 125 kHz carrier frequency and a 125 kHz communication bandwidth, using data packets of 125 kilobytes and control packets at 64 kbps. For what concerns the OFDM communication. The available bandwidth is equally split among $N = 8$ OFDM subcarriers, while the transmitting node can select $K = 3$ subcarriers among the 8 available. In particular, we configured subcarriers 3, 5, and 8 to exhibit favorable channel conditions, while we set the remaining five subcarriers to experience poor channel quality and/or substantial interference. Accordingly, selecting the high-quality subcarriers 3, 5, and 8 represents the optimal choice in our scenario. Notably, this choice yields a maximum PDR of 45.71%, as determined from simulation-based evaluation.

Each episode duration was set to $\Delta = 300$ seconds. This duration ensures a favorable trade-off between responsiveness to channel variations and computational efficiency. It allows the agent to capture meaningful changes in UW conditions without incurring excessive decision noise or overhead, thereby supporting stable and effective subcarrier adaptation throughout all simulations.

Regarding the MERMAID agent, we set the exploration/exploitation balance parameter c to 2. The agent is implemented in Python and interacts with the DESERT simulator through dedicated coordination scripts, which handle real-time exchange of link statistics and dynamic reconfiguration of transmission parameters.

All learning-based methods, including MERMAID and the DRL baselines, were executed on a workstation with dual NVIDIA GeForce RTX3070 GPUs. On average, each training phase—including interaction with the DESERT simulator—required approximately 90 minutes.

B. Convergence Analysis

Fig. 2 shows the cumulative episode reward achieved by the proposed MERMAID framework, benchmarked against the DDQN and PPO baselines.

MERMAID quickly converges to the maximum cumulative reward within just a few episodes. In contrast, the DRL-based baselines exhibit oscillatory behavior and slower convergence, highlighting their difficulty in balancing exploration and exploitation. These instabilities are typical of DRL algorithms, which require extensive training to stabilize. While MERMAID rapidly locks onto an effective subcarrier configuration, both DDQN and PPO continue to explore suboptimal actions, leading to higher variability and slower performance improvements in the early stages.

Among the DRL baselines, DDQN slightly outperforms PPO—an outcome consistent with their respective learning mechanisms. DDQN is a value-based algorithm that mitigates the overestimation bias of standard Q-learning by decoupling action selection from action evaluation via a target network. This design allows it to make more stable decisions based on link statistics, refined over time through experience replay. PPO, by contrast, is a policy-gradient algorithm that relies on clipped surrogate objectives and adaptive trust region constraints to learn a stochastic subcarrier selection policy. While PPO excels in scenarios with continuous action spaces and interference-driven dynamics, its performance in underwater OFDM settings is limited by the high variance in gradient estimation and the absence of external interferers. Overall, MERMAID surpasses both DRL methods by converging more

rapidly and providing reliable subcarrier adaptation with minimal training overhead.

MERMAID's superiority stems from the intrinsic advantages of MAB algorithms. Unlike DRL methods, which depend on large datasets, deep networks, and lengthy training, MABs offer a lightweight, stateless approach that quickly identifies promising actions from immediate feedback. This is particularly effective in settings with finite and structured action spaces, like adaptive subcarrier selection in UW-OFDM. By focusing exploration on underused but potentially rewarding configurations and promptly exploiting high-performing ones, MERMAID ensures faster convergence and greater stability. Moreover, MABs are more sample-efficient than DRL, requiring fewer interactions to approach optimality, making them ideal for resource-constrained IoUT devices.

C. Packet Delivery Ratio Analysis

To assess communication reliability, we analyze the average PDR achieved during simulation. Fig. 3 compares MERMAID with the two DRL-based baselines, highlighting each approach's ability to maintain high delivery performance under dynamic channel conditions.

The results show that MERMAID consistently outperforms both DDQN and PPO in average PDR. Specifically, it achieves **9.35% higher PDR than DDQN** and **21.15% higher than PPO**. Moreover, it almost matches the maximum PDR achievable on the channel, i.e., 45.71%. This performance gain stems from MERMAID's ability to dynamically adapt the active subcarrier subset based on real-time feedback, enabling more responsive decisions. By promptly reacting to link quality fluctuations, MERMAID reduces packet loss and improves overall transmission reliability.

In contrast, DRL-based approaches exhibit delayed convergence and higher variance in early training phases, often resulting in suboptimal subcarrier allocations. DDQN, while relatively stable, requires numerous episodes to stabilize its Q-value estimates. PPO, which relies on stochastic policy gradients, shows greater instability and poorer performance under interference-prone conditions, confirming its limited suitability for deterministic environments such as OFDM-based UWA links.

In summary, the results confirm MERMAID's robustness in ensuring reliable communication under challenging UWA conditions, making it a strong candidate for adaptive subcarrier selection in IoUT systems.

D. Energy Efficiency Analysis

We evaluate MERMAID's energy efficiency by measuring the average energy required to successfully deliver a packet. Fig. 4 compares MERMAID with the DRL-based baselines DDQN and PPO. This metric is crucial in UW networks, where energy resources are limited and recharging opportunities are often unavailable.

The plot clearly shows that MERMAID consistently achieves the highest energy efficiency, **reducing consumption by 18.50% over DDQN and 45.40% over PPO**. This

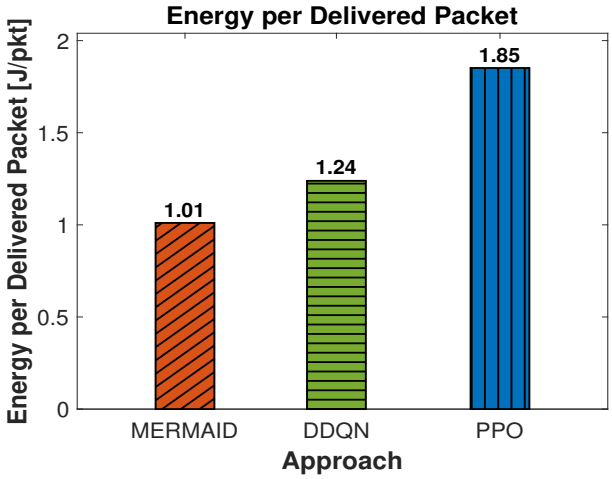


Fig. 4: Energy consumption comparison between MERMAID, DDQN and PPO.

No. of high-quality subcarriers selected	**MERMAID**	**DDQN**	**PPO**
0	1.0%	5.2%	18.1%
1	3.0%	29.2%	54.8%
2	1.5%	34.5%	25.5%
3	94.5%	31.1%	1.6%

TABLE I: Percentage of times each approach selects a given number of high-quality subcarriers.

improvement derives from the CMAB model's lightweight structure and focused decision-making, which avoids the long trial-and-error cycles of deep learning models.

While DDQN and PPO rely on prolonged exploration and extensive training—often yielding inconsistent decisions and wasted energy—MERMAID's streamlined learning quickly identifies optimal configurations with minimal overhead. This enables faster stabilization and reduced transmission costs, making it ideal for energy-constrained IoUT devices.

In conclusion, the results confirm MERMAID's ability to ensure robust communication and extended operational lifetime, both essential for sustainable, long-term deployment in UW environments.

E. Subcarrier Selection Strategy

To assess subcarrier selection accuracy under interference, Tab. I shows the percentage of decision episodes in which MERMAID, DDQN, and PPO correctly identified 0, 1, 2, or all 3 high-quality subcarriers. This metric reflects each agent's ability to detect favorable transmission opportunities across the OFDM spectrum.

The results clearly highlight MERMAID's superiority in accurate and consistent decision-making. It correctly identifies all three interference-free subcarriers in 94.5% of the episodes, demonstrating strong adaptability to dynamic channel conditions. Incorrect or partial selections are rare, confirming MERMAID's ability to learn and exploit the spectral structure of UW environments.

By contrast, DDQN correctly selects all three interference-free subcarriers in only 31.1% of the episodes. More often, it identifies just one or two, indicating a less stable learning process and a weaker understanding of the spectral environment. While it occasionally makes accurate decisions, its performance is significantly more variable than MERMAID's.

PPO, on the other hand, exhibits significantly lower accuracy, correctly identifying all interference-free subcarriers in only 1.6% of the episodes. In over 72% of cases, it selects none or only one, leading to inefficient transmissions and increased packet loss.

These results further confirm the effectiveness of the proposed CMAB-based framework in enabling high-precision spectrum access. MERMAID's consistent identification of optimal subcarriers enhances both reliability and throughput, while reducing energy waste caused by transmissions over impaired channels—underscoring its suitability for energy-constrained and interference-prone UW deployments.

V. Conclusions

This paper presented MERMAID, a Combinatorial Multi-Armed Bandit (CMAB)-based framework for adaptive subcarrier selection in Orthogonal Frequency Division Multiplexing (OFDM)-based UnderWater Acoustic (UWA) networks. MERMAID is the first CMAB-based framework to tackles the key challenge of OFDM carrier selection in Underwater Acoustic channels in real time.

Comprehensive simulations on the DESERT framework confirmed MERMAID's effectiveness across multiple performance metrics. In particular, we compared MERMAID against two Deep Reinforcement Learning baselines, namely DDQN and PPO. MERMAID outperforms DDQN and PPO methods by achieving up to **9.35%** and **21.15%** higher packet delivery ratio (PDR), respectively. Moreover, MERMAID also reduces the average energy per delivered packet by 18.50% with respect to DDQN and 45.40% relative to PPO. The main reason behind MERMAID's performance lies in its superior subcarrier selection accuracy; MERMAID correctly identifies all high-quality subcarriers in over 94% of the cases —vastly outperforming DRL baselines, which suffer from instability and frequent selection errors.

The convergence analysis confirms that MERMAID attains optimal behavior sensibly faster than DRL agents, avoiding the extended training and high variance typical of neural methods. Thanks to its lightweight design and low complexity, MERMAID proves especially suitable for resource-constrained, interference-prone IoUT scenarios.

In summary, MERMAID offers a robust, adaptive, and energy-efficient solution for subcarrier selection in UWA networks, enabling the deployment of intelligent communication systems in next-generation applications.

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